Biases are the prior knowledge we have about the structure of the data we wish to learn, and they make learning much faster. But there's more to it than just biases. With a convolutional neural network (CNN), the bias is toward learning local features, but CNNs also have the property of translation *invariance*. A function is said to be invariant to a particular transformation of its input when such a transformation does not change the output. For example, the addition function is invariant to the order of its inputs $add(x,y) = add(y,x)$, whereas the subtraction operator does not share this order invariance (this particular invariant property has its own special name: *commutativity*). In general, a function, $f(x)$, is invariant with respect to some transformation, $g(x)$, to its input, x, when $f(g(x)) = f(x)$. CNNs are functions in which a translation (movement up, down, left, or right) of an object in an image will not impact the behavior of the CNN classifier; it is invariant to translation (top panel of figure 10.3).

Figure 10.3. Invariance: Rotational invariance is a property of the function such that a rotation transformation of the input does not change the output of the function. Equivariance: Translational equivariance for a function is when applying the translation to the input results in the same output as when you apply the translation after the function has already been performed on the unaltered input.